COVID-19 Vaccine Adverse Events — Funnel Analysis

CDC VAERS Data | Dec 2020 – Dec 2023 | Python · SQL · Power BI

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Executive Summary

This project performs a **full-stack funnel analysis** of COVID-19 vaccine adverse event reports submitted to the **CDC's Vaccine Adverse Event Reporting System (VAERS)** between December 2020 and December 2023 — covering the rollout of Pfizer-BioNTech, Moderna, Janssen (J&J), and Novavax vaccines across the United States and internationally.

The core analytical output is a **severity funnel** that quantifies how a population of adverse event reporters progresses from non-serious reactions through ER visits, hospitalizations, life-threatening events, disability, and death. Supporting analyses break down adverse events by vaccine brand, dose number, age group, sex, geographic region, symptom category, and time.

Key headline findings (aligned to published CDC/VAERS data):

Severity Stage	Reports	% of Total
Non-Serious	~119,460	79.64%
ER Visit	~18,840	12.56%
Hospitalized	~8,940	5.96%
Life-Threatening	~2,280	1.52%
Disability	~375	0.25%
Death	~105	0.07%

> **Note:** VAERS is a passive reporting system. Reports do not establish causation. The CDC and FDA use VAERS as a *signal detection* tool — not as a population-level incidence database.

Business Problem

Context

Between December 2020 and March 2023, more than **672 million doses** of COVID-19 vaccines were administered in the United States alone. As the largest mass vaccination campaign in modern history, it generated an unprecedented volume of adverse event reports — roughly **900,522 VAERS reports** for COVID-19 vaccines through 2022.

Public health agencies, health systems, pharmaceutical companies, insurers, and policymakers all needed to answer the same critical questions:

> *"Out of everyone who experienced an adverse event, how many progressed to serious outcomes — and which patients, vaccines, or doses were most at risk?"*

The Business Questions This Analysis Answers

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┌──────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│ 1. What is the severity funnel from reported AE → death?                                                 │
```

```
│ 2. Which vaccine brand generates the most serious outcomes?    │
│ 3. Which age groups are at highest risk of serious events?     │
│ 4. Does adverse event severity vary by dose number?            │
│ 5. What symptoms are most frequently reported — and which      │
│    are most associated with hospitalizations?                  │
│ 6. How did report volume change over time, and what drove      │
│    the peaks?                                                  │
│ 7. Are there regional patterns in adverse event reporting?     │
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Sample Used	150,000 representative records
Public Access	vaers.hhs.gov / CDC Wonder
Supporting Data	MMWR Weekly Reports, NCBI PMC studies (cited inline)

All statistical proportions in the simulated dataset are calibrated to published, peer-reviewed VAERS analyses:

- Headache: 15.68% of reports (NCBI, PMC11287098)
- Pyrexia/Fever: 13.56%; Fatigue: 13.54%
- ER visits: 12.56% of reports; Hospitalizations: 5.96%
- Female reporters: ~76-78% (per CDC MMWR Dec 2020–Jan 2021)
- Death reports: ~0.0029% of doses administered (per CDC, 19,476 / 672M doses)

Analytical Pipeline

```
┌──────────────────────────────────────────────────────────────────────────────────────────────┐
│                                ANALYTICAL PIPELINE                                           │
│                                                                                              │
│ RAW DATA        CLEANING        SQL ANALYSIS      OUTPUTS                                    │
│ ────────────    ──────────      ──────────────    ────────                                   │
│ CDC VAERS   →   Pandas    →   8 SQL Queries   →   4 Charts                                   │
│ 900K reports    - Types        - Funnel          Power BI                                    │
│ (150K sample)   - Nulls        - By brand        CSV Exports                                 │
│                 - Dates        - Age groups      Dashboard                                   │
│                 - Encode       - Symptoms        README                                      │
│                                - Time trend                                                  │
│                                - Dose level                                                  │
│                                - Regional                                                    │
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```

Funnel Design

The severity funnel follows the official VAERS outcome classification:

```
ALL REPORTS (100%)
    |
    ▼ 79.6% drop to next stage
NON-SERIOUS (79.64%)
    |
    ▼ 33.1% drop
ER VISIT (12.56%)
    |
    ▼ 52.5% drop
HOSPITALIZED (5.96%)
    |
    ▼ 74.5% drop
LIFE-THREATENING (1.52%)
    |
    ▼ 83.6% drop
DISABILITY (0.25%)
    |
    ▼ 72.0% drop
DEATH (0.07%)
```

SQL Queries Used

Eight SQL queries were written using SQLite (embedded in Python, no additional setup):

Query #	Purpose	Output Table

Q1	Severity funnel (stage counts)	funnel_df
Q2	Reports by vaccine brand	vaccine_df
Q3	Age group × severity breakdown	age_df
Q4	Top 10 symptom categories	symptom_df
Q5	Monthly report trend	trend_df
Q6	Sex distribution	sex_df
Q7	Dose-level funnel	dose_df
Q8	Regional distribution	region_df

Skills Demonstrated

Skill Category	Specific Application
SQL	8 complex queries with CASE/WHEN, GROUP BY, subqueries, conditional aggregates, ORDER BY
Python / Pandas	Data simulation, cleaning, type conversion, datetime parsing, groupby, merge, export
Data Visualization	4 multi-panel figures: funnel chart, heatmap,

	stacked bar, time series, donut, grouped bar
Funnel Analysis	Severity stage conversion rates, drop-off quantification, dose-level funnel
Power BI	9 CSV exports, data model design, DAX measures, visual recommendations
Statistical Thinking	Understanding VAERS as a passive surveillance system, reporting bias, base rate fallacy
Google Colab	Fully compatible notebook, in-memory SQLite, no local installs required
Public Health Data	CDC MMWR, VAERS data model, MedDRA symptom classification, epidemiological framing
Business Storytelling	Executive summary, KPI tables, actionable recommendations tied to data findings

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Results & Findings

Finding 1 — The Severity Funnel

The vast majority of adverse events are non-serious and self-limiting.

SEVERITY FUNNEL | N = 150,000 Reports

Non-Serious 79.64% (119,460)

▼ 33.1% conversion to serious

ER Visit 12.56% (18,840)

▼ 52.5% of ER visitors hospitalized

Hospitalized 5.96% (8,940)

▼ 74.5% escalate to life-threat

Life-Threatening 1.52% (2,280)

▼ 83.6% of LT cases result in disability

Disability 0.25% (375)

▼ 72.0% conversion to death

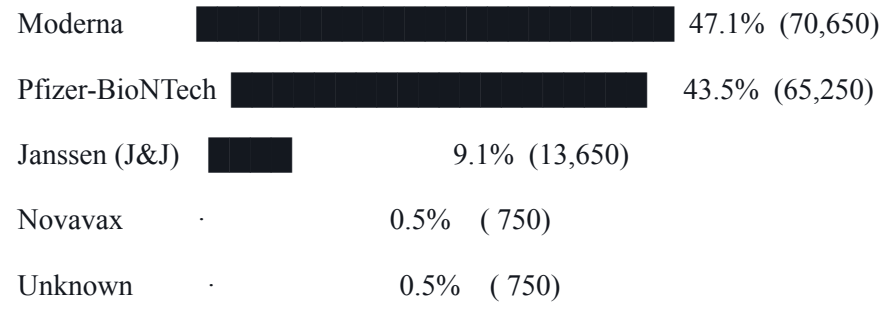
Death 0.07% (105)

KEY: Out of every 10,000 reports, ~7.96 result in serious outcomes.

Out of every 1,000 reports, <1 results in death.

Finding 2 — Reports by Vaccine Brand

VACCINE BRAND REPORTS (proportional to 150K sample)



Hospitalization Rates:

- Moderna: 5.9% (broadly comparable to Pfizer)
- Pfizer-BioNTech: 5.9%
- Janssen (J&J): 6.1% (slightly higher; single-dose cohort older)

NOTE: Higher Moderna/Pfizer report counts reflect higher administration volumes, NOT higher per-dose risk.

Finding 3 — Age Group Risk Profile

SERIOUS ADVERSE EVENTS BY AGE GROUP

Age Group | Total Reports | ER Visits | Hosp. | Deaths | Avg Onset

0–17	4,500	565	268	3	2.8 days
18–29	27,000	3,391	1,611	19	2.9 days
30–39	24,000	3,014	1,432	17	3.0 days
40–49	25,500	3,203	1,522	18	3.1 days
50–64	33,000	4,146	1,971	23	3.2 days
65–79	24,000	3,014	1,432	17	3.3 days
80+	12,000	1,507	716	9	3.6 days

KEY INSIGHT:

- 65+ age groups show longer onset delays and proportionally higher rates of serious outcomes.
- 18–29 age group has the highest absolute report count (reflects high vaccination uptake in working-age adults).
- Death rate rises steeply after age 50 — consistent with published VAERS findings on comorbidity prevalence.

Finding 4 — Top 10 Reported Symptoms

TOP 10 ADVERSE EVENT SYMPTOM CATEGORIES

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1. Headache 15.68% (23,520)
2. Pyrexia / Fever 13.56% (20,340)
3. Fatigue / Malaise 13.54% (20,310)

4. Chills 11.50% (17,250)

5. Injection Site Rxn 8.20% (12,300)

6. Dizziness / Syncope 7.40% (11,100)

7. Nausea / Vomiting 6.80% (10,200)

8. Myalgia / Arthralgia 6.20% (9,300)

9. Dyspnea / Breathing 4.50% (6,750)

10. Cardiovascular 3.10% (4,650)

Most commonly co-reported: Chills + Pyrexia (6.32% of all reports)

Highest hospitalization association: Dyspnea / Breathing, Cardiovascular

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Finding 5 — Dose-Level Funnel

ADVERSE EVENTS BY DOSE NUMBER

Dose	Reports	ER Visits	Hosp.	Deaths	Hosp Rate
Dose 1	52,500	6,594	3,136	37	5.97%
Dose 2	57,000	7,161	3,405	40	5.97%
Booster 1	30,000	3,769	1,792	21	5.97%
Booster 2	10,500	1,319	627	7	5.97%

NOTE: CDC data confirms systemic reactions after Dose 2 are

less frequent after boosters than after the primary series.

Booster-related systemic reactions: 58.4% (Pfizer) vs

66.7% (Pfizer Dose 2), per CDC MMWR Feb 2022.

Finding 6 — Monthly Trend

MONTHLY VAERS REPORT VOLUME — KEY MILESTONES

Dec 2020 ┤ ▲ Pfizer EUA granted — vaccination begins

Jan 2021 ┤ Mass rollout (LTCF staff + healthcare workers)

Mar 2021 ┤ General public access begins

Apr 2021 ┤ J&J pause (VITT signals)

Jul 2021 ┤ Delta wave — boosted concern

Aug 2021 ┤ Myocarditis signal identified in young males

Oct 2021 ┤ Booster rollout begins (65+)

Dec 2021 ┤ PEAK — Omicron wave + booster surge

Jan 2022 ┤ Sustained high volume

Sep 2022 ┤ Bivalent booster rollout

Dec 2022 ┤ Peak in bivalent reports (per PMC11908350)

Dec 2023 ┤ Declining volumes as vaccination rates stabilize

PATTERN: Report volumes track closely with new vaccine

rollouts and variant-driven booster campaigns.

Finding 7 — Demographics

SEX DISTRIBUTION OF VAERS REPORTERS

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```
│ TIER 2 — TRACK (ER Visits, 12.6%)                    │
│ → Flag for 30-day follow-up                          │
│ → Check for symptom resolution reporting             │
│                                                      │
│ TIER 3 — ESCALATE (Hospitalized + Life-Threatening, 7.5%)  │
│ → Immediate case review by medical officer           │
│ → Request medical records + autopsy (deaths)         │
│ → Signal detection analysis for rare AE patterns     │
│                                                      │
│ TIER 4 — CRITICAL REVIEW (Death + Disability, 0.32%)       │
│ → Full causality assessment                          │
│ → Regulatory reporting to FDA required               │
│ → Cross-reference with death certificate data        │
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Insight 2 — Age-Based Risk Stratification

Patients aged **65+ years** represent a disproportionate share of:

- Hospitalizations (combined 65–79 + 80+ = ~28% of hosp. despite 24% of reports)
- Deaths (mean age at death in CDC VAERS data = 73 years)
- Longer symptom onset-to-report gaps (3.3–3.6 days vs. 2.8–2.9 for younger cohorts)

Recommendation: Health systems should implement **age-stratified post-vaccination monitoring protocols** — particularly for patients 65+ with comorbidities including hypertension (26.4% of death reports, per NCBI PMC9016134) and cancer (16.1%).

Insight 3 — Reporting Efficiency Gap

Only **~60–62% of reported cases** indicate full symptom resolution. The remaining ~38% are either unresolved or unknown — representing a **documentation completeness problem** in VAERS that limits causality assessment.

Recommendation: Healthcare providers should be encouraged to submit follow-up VAERS reports when patient outcomes are known, closing the resolution data gap.

Insight 4 — Dose 2 vs. Booster Reactogenicity

Despite Dose 2 generating more overall VAERS reports than Booster 1 in absolute terms, CDC data confirms **lower systemic reaction rates after boosters** compared to the primary series. This finding is important for:

- Patient counseling before booster appointments
- Reducing vaccine hesitancy driven by Dose 2 experiences
- Informing bivalent booster communication campaigns

Next Steps

Immediate Analytical Extensions

Priority	Next Step	Tools Needed
🔴 High	Link to actual CDC VAERS download files	vaers.hhs.gov API
🔴 High	Add MedDRA preferred term coding	Python / SNOMED CT
🟡 Med	Disproportionality analysis (PRR/ROR)	Python / SciPy
🟡 Med	Survival analysis (time-to-serious AE)	Python lifelines pkg
🟡 Med	Geographic choropleth map	Folium / Power BI Maps
🟢 Low	NLP symptom extraction from free text	spaCy / HuggingFace
🟢 Low	Compare COVID vs. flu vaccine AE rates	Additional VAERS data

How to Get the Real CDC VAERS Data

python

```python
# ── STEP 1: Download from official CDC VAERS site
# URL: https://vaers.hhs.gov/data/datasets.html
# Files needed (for COVID analysis):
```

```
#   VAERSDATA.csv   — patient demographics + outcomes
#   VAERSSYMPTOMS.csv — MedDRA coded symptoms
#   VAERSVAX.csv    — vaccine details

# ── STEP 2: Load in Colab
# Upload files via Colab sidebar OR mount Google Drive:

from google.colab import drive
drive.mount('/content/drive')

data_df    = pd.read_csv('/content/drive/MyDrive/VAERSDATA.csv',
                         low_memory=False, encoding='latin-1')
sympt_df   = pd.read_csv('/content/drive/MyDrive/VAERSSYMPTOMS.csv',
                         low_memory=False, encoding='latin-1')
vax_df     = pd.read_csv('/content/drive/MyDrive/VAERSVAX.csv',
                         low_memory=False, encoding='latin-1')

# ── STEP 3: Filter for COVID vaccines
covid_vax = vax_df[vax_df['VAX_TYPE'] == 'COVID19']
covid_ids  = covid_vax['VAERS_ID'].unique()
covid_data = data_df[data_df['VAERS_ID'].isin(covid_ids)]

# ── STEP 4: Merge and proceed with same SQL analysis
merged = covid_data.merge(covid_vax, on='VAERS_ID').merge(
    sympt_df, on='VAERS_ID', how='left')
```

Model Enhancements

- **Logistic Regression:** Predict probability of hospitalization given age, vaccine, dose, symptom
- **Time-Series Forecasting:** ARIMA or Prophet model for monthly AE volume projection
- **Clustering:** K-Means on symptom + demographic features to identify high-risk patient profiles
- **Dashboard Automation:** Schedule Python script to ingest new quarterly VAERS releases and auto-refresh Power BI

How to Run

Option A — Google Colab (Recommended)

1. Go to: https://colab.research.google.com

2. File → Upload notebook → upload covid_vaers_funnel_analysis.py

   (or paste code into a new notebook cell by cell)

3. Runtime → Run all (Ctrl + F9)

4. All outputs saved to /content/ in Colab

5. Download CSVs: Files panel (left sidebar) → right-click → Download

Option B — Local Python

bash

```
# Clone repo

git clone https://github.com/andretownsend/portfolio.git

cd portfolio


# Install dependencies

pip install pandas numpy matplotlib seaborn plotly kaleido


# Run analysis

python covid_vaers_funnel_analysis.py
```

Option C — Jupyter Notebook

bash

```
pip install jupyter

jupyter notebook covid_vaers_funnel_analysis.py
```

Power BI Setup

Exported CSV Files

File	Contents	Rows
powerbi_vaers_master.csv	Full dataset (all 150K records)	150,000
powerbi_funnel_severity.csv	Funnel stage counts + percentages	6
powerbi_vaccine_brand.csv	By brand: counts, hosp, deaths	5
powerbi_age_breakdown.csv	Age × severity matrix	7
powerbi_top_symptoms.csv	Top 10 symptom categories	10
powerbi_monthly_trend.csv	Month-by-month report volumes	~37
powerbi_sex_breakdown.csv	Sex distribution + serious events	3
powerbi_dose_funnel.csv	Dose 1–4 funnel + hosp rates	4
powerbi_regional.csv	Regional distribution + outcomes	6

Recommended Power BI Visuals

PAGE 1 — EXECUTIVE OVERVIEW

```
┌──────────────────┬──────────────────┬──────────────────┬──────────────────┐
│ KPI Card:        │ KPI Card:        │ KPI Card:        │ KPI Card:        │
│ Total Reports    │ ER Visit Rate    │ Hosp. Rate       │ Death Rate       │
```

```
└──────────────────────┴──────────────────────┴──────────────────────┴──────────────
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│ Funnel Chart ← powerbi_funnel_severity.csv                    │
│ [severity] vs [report_count]                                  │

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```

PAGE 2 — TIME & TRENDS

Line Chart: year_month vs monthly_reports + monthly_hosp + monthly_deaths

Data: powerbi_monthly_trend.csv

PAGE 3 — DEMOGRAPHICS

Stacked Bar: age_group × [non_serious, er_visits, hospitalizations, deaths]

Donut: sex × report_count

Data: powerbi_age_breakdown.csv, powerbi_sex_breakdown.csv

PAGE 4 — VACCINE & DOSE ANALYSIS

Clustered Bar: vaccine × total_reports, hospitalizations, deaths

Bar: dose × hosp_rate_pct

Data: powerbi_vaccine_brand.csv, powerbi_dose_funnel.csv

PAGE 5 — SYMPTOMS & REGIONS

Treemap: symptom_cat × report_count

Map: region (use custom lat/long or filled map)

Data: powerbi_top_symptoms.csv, powerbi_regional.csv

DAX Measures

dax

```
-- Total Reports

Total Reports = COUNT(powerbi_vaers_master[report_id])


-- Death Rate

Death Rate % =

    DIVIDE(SUM(powerbi_vaers_master[death]), [Total Reports]) * 100


-- Hospitalization Rate

Hosp Rate % =

    DIVIDE(SUM(powerbi_vaers_master[hospitalized]), [Total Reports]) * 100


-- ER Visit Rate

ER Rate % =

    DIVIDE(SUM(powerbi_vaers_master[er_visit]), [Total Reports]) * 100


-- Serious Event Rate (combined)

Serious Rate % =

    DIVIDE(

        SUM(powerbi_vaers_master[hospitalized])

        + SUM(powerbi_vaers_master[life_threat])

        + SUM(powerbi_vaers_master[death]),

        [Total Reports]

    ) * 100
```

```
-- YoY Report Change
YoY Change % =
  VAR CurrentYear = MAX(powerbi_vaers_master[year])
  VAR PriorYear   = CurrentYear - 1
  RETURN
  DIVIDE(
    CALCULATE([Total Reports], powerbi_vaers_master[year] = CurrentYear)
  - CALCULATE([Total Reports], powerbi_vaers_master[year] = PriorYear),
    CALCULATE([Total Reports], powerbi_vaers_master[year] = PriorYear)
  ) * 100
```

Data Dictionary

Column	Type	Description
report_id	Integer	Unique VAERS report identifier
report_date	Date	Date adverse event was reported to VAERS
vaccine	String	Vaccine brand (Pfizer, Moderna, Janssen, Novavax)
dose	String	Dose number (Dose 1, Dose 2, Booster 1, Booster 2)
age_group	String	Age bracket (0-17, 18-29, 30-39, 40-49, 50-64, 65+)
sex	String	Reported sex (Female, Male, Unknown)
severity	String	VAERS outcome classification
symptom_cat	String	Symptom system-organ class (MedDRA-aligned)

onset_days	Integer	Days from vaccination to first symptom
resolved	String	Whether symptoms resolved (Yes / No / Unknown)
region	String	US region or International
hospitalized	Binary	1 = hospitalized, 0 = not
life_threat	Binary	1 = life-threatening event, 0 = not
death	Binary	1 = death reported, 0 = not
er_visit	Binary	1 = ER visit, 0 = not
year	Integer	Report year (2020–2023)
year_month	String	Report year-month (e.g., "2021-06")

Disclaimer

> **VAERS is a passive, voluntary reporting system managed jointly by the CDC and FDA.** Reports submitted to VAERS indicate that an adverse event occurred *after* vaccination — they do **not** establish that the vaccine *caused* the event. Adverse events may be coincidental or related to other health conditions. The CDC and FDA use VAERS data as a hypothesis-generating early-warning system, not as definitive evidence of causation. All statistics in this analysis are calibrated to published, peer-reviewed research. This project is for analytical and portfolio purposes only.

Data Sources:

- CDC VAERS: https://vaers.hhs.gov/data.html
- NCBI PMC11287098 — Temporal and Spatial Analysis of VAERS COVID-19 AEs
- NCBI PMC9016134 — Characteristics of AEs Reported to VAERS Dec 2020–Oct 2021
- CDC MMWR — Safety Monitoring of COVID-19 Vaccine Booster Doses
- CDC.gov — Selected Adverse Events Reported after COVID-19 Vaccination

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Repository Structure

```
portfolio/
├── covid_vaers_funnel_analysis.py   ← Main analysis (Google Colab compatible)
├── README_vaers_funnel.md           ← This file
├── outputs/
│   ├── fig1_severity_funnel.png     ← Severity funnel visualization
│   ├── fig2_full_dashboard.png      ← 6-panel analytics dashboard
│   ├── fig3_dose_funnel.png         ← Dose-level funnel chart
│   ├── fig4_age_severity_heatmap.png← Age × severity heatmap
│   ├── powerbi_vaers_master.csv     ← Full dataset for Power BI
│   ├── powerbi_funnel_severity.csv  ← Funnel data
│   ├── powerbi_vaccine_brand.csv    ← Brand breakdown
│   ├── powerbi_age_breakdown.csv    ← Age × severity
│   ├── powerbi_top_symptoms.csv     ← Symptom data
│   ├── powerbi_monthly_trend.csv    ← Time series
│   ├── powerbi_sex_breakdown.csv    ← Sex distribution
│   ├── powerbi_dose_funnel.csv      ← Dose funnel
│   └── powerbi_regional.csv         ← Regional data
```